* $S^{\perp} = \{ v \in V : \langle v, u \rangle = 0 \ \forall \ u \in S \}$

let $v_1, v_2 \in S^{\perp}$

$\therefore \langle v_1, u \rangle = 0 \ , \langle v_2, u \rangle = 0 \quad \forall u \in S.$

$$\langle v_1 + v_2 , u \rangle = \langle v_1, u \rangle + \langle v_2, u \rangle$$
$$= 0 + 0$$
$$= 0$$

$\therefore v_1 + v_2 \in S^{\perp}$

Now, $\langle \alpha v_1, u \rangle = \alpha \langle v_1, u \rangle$

$= \alpha \cdot 0$

$= 0$

$\therefore \alpha v_1 \in S^{\perp}$

$\therefore S^{\perp}$ is a subspace of V. □

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$S = \{ u_1, u_2, \ldots, u_n \}$

$v = \alpha_1 u_1 + \alpha_2 u_2 + \ldots + \alpha_n u_n$

$\langle v, u_i \rangle = \langle \alpha_1 u_1 + \alpha_2 u_2 + \ldots + \alpha_n u_n, u_i \rangle$

$= \alpha_1 \langle u_1, u_i \rangle + \alpha_2 \langle u_2, u_i \rangle + \ldots + \alpha_n \langle u_n, u_i \rangle$

$\langle u_i, u_j \rangle = 0 \ , \ i \neq j$

$= \alpha_i \langle u_i, u_i \rangle$

$\Rightarrow \alpha_i = \dfrac{\langle v, u_i \rangle}{\langle u_i, u_i \rangle}$, $i = 1, 2, \ldots, n$, where α_i's are the fourier co-efficients w.r.t u

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